

LAB SESSION 8

Lab Name: Verification, Validation & Software Testing

Project: MediScript-E — Digital Healthcare Platform

Objectives

- Ensure correctness of MediScript-E through systematic testing
- Design comprehensive test cases for all major modules
- Create a Requirement Traceability Matrix (RTM)

Theory

- **Verification:** Building the system right — ensuring implementation matches design and specifications
- **Validation:** Building the right system — ensuring the system meets actual user needs

Testing improves confidence, not perfection. For MediScript-E, testing covers authentication, appointment management, prescription handling, medical vault, and admin operations.

Task 1: Test Plan

1.1 Test Scope

The following modules are covered:

- Authentication (Registration, Login, Email Verification, OAuth, 2FA)
- Patient Module (Appointments, Prescriptions, Medicine Reminders, Medical Vault)
- Doctor Module (Appointment Management, Prescription Issuance)
- Admin Module (Dashboard, User Management, Appointment Overview)
- Settings Module (Profile Update, Password Change, 2FA Toggle)

1.2 Test Types

Test Type	Description	Tool/Method
Unit Testing	Test individual API routes and functions	Manual / Jest
Integration Testing	Test interaction between modules	Manual API testing
Functional Testing	Test against functional requirements	Browser-based manual testing
Security Testing	Test authentication and authorization	Manual penetration testing
UI Testing	Test responsiveness and user interactions	Browser DevTools

1.3 Test Environment

Environment	Details
Local	<code>http://localhost:3000</code> with <code>.env</code> configuration
Production	<code>https://mediscript-e.vercel.app</code>
Database	Supabase PostgreSQL (same DB for both)
Browser	Chrome, Firefox, Safari

1.4 Entry and Exit Criteria

Criteria	Details
Entry	All features implemented, build successful (<code>pnpm run build</code>)
Exit	All High priority test cases pass, no critical bugs

Task 2: Test Cases

Module 1: Authentication

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-01	Register with valid credentials (Patient)	Valid name, email, password, role=PATIENT, bloodGroup=O+	Account created, verification email sent, success message shown	High	Pass
TC-02	Register with valid credentials (Doctor)	Valid name, email, password, role=DOCTOR, licenseNo, specialization	Account created, verification email sent	High	Pass
TC-03	Register with existing email	Already registered email	Error: "Email already registered"	High	Pass
TC-04	Register with invalid email format	<code>notanemail</code>	Validation error shown	High	Pass
TC-05	Register with short password	Password < 6 characters	Error: "Password must be at least 6 characters"	High	Pass
TC-06	Verify email with valid token	Valid token from email	<code>emailVerified: true</code> , redirect to login	High	Pass

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-07	Verify email with expired token	Token older than 24 hours	Error: "Verification link expired"	High	Pass
TC-08	Login with valid credentials	Verified email + correct password	JWT session created, redirect to /dashboard	High	Pass
TC-09	Login with wrong password	Valid email + wrong password	Error: "Invalid Email or Password"	High	Pass
TC-10	Login with unverified email	Unverified account credentials	Error: "Please verify your email" + resend option	High	Pass
TC-11	Login with Google OAuth	Valid Google account	Account created (if new), redirect to /dashboard	Medium	Pass
TC-12	Login with GitHub OAuth	Valid GitHub account	Account created (if new), redirect to /dashboard	Medium	Pass
TC-13	Enable 2FA from settings	Toggle 2FA on	twoFactorEnabled: true saved in database	High	Pass
TC-14	Login with 2FA enabled	Valid credentials + 2FA enabled	OTP sent to email, redirect to /verify-2fa	High	Pass
TC-15	Verify OTP with correct code	Valid 6-digit OTP	Session created, redirect to /dashboard	High	Pass
TC-16	Verify OTP with wrong code	Invalid OTP	Error: "Invalid OTP"	High	Pass
TC-17	Verify OTP after expiry	OTP older than 10 minutes	Error: "OTP has expired"	High	Pass
TC-18	Resend OTP	Click resend after 60 seconds	New OTP sent, countdown resets	Medium	Pass
TC-19	Direct 2FA bypass attempt	POST signIn with __2fa_verified__ without OTP	Error: "Invalid credentials"	High	Pass

Module 2: Patient — Appointments

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-20	Book appointment with valid data	Doctor, date, time, reason	Appointment created with status PENDING	High	Pass
TC-21	Book appointment without selecting doctor	No doctor selected	Validation error	High	Pass
TC-22	View all appointments	Logged in as patient	All patient appointments displayed with status	High	Pass
TC-23	Cancel pending appointment	Appointment with status PENDING	Status updated to CANCELLED	Medium	Pass

Module 3: Doctor — Appointments & Prescriptions

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-24	View assigned appointments	Logged in as doctor	All doctor appointments displayed	High	Pass
TC-25	Confirm pending appointment	Appointment with status PENDING	Status updated to CONFIRMED	High	Pass
TC-26	Cancel appointment	Any active appointment	Status updated to CANCELLED	High	Pass
TC-27	Complete confirmed appointment	Appointment with status CONFIRMED	Status updated to COMPLETED	High	Pass
TC-28	Issue prescription with valid data	Valid patientId, diagnosis, medications	Prescription created, visible to patient	High	Pass
TC-29	Issue prescription with invalid patient ID	Non-existent patientId	Error: "Patient not found"	High	Pass
TC-30	Issue prescription with missing fields	Empty diagnosis or medications	Validation error	High	Pass

Module 4: Patient — Prescriptions & Medical Vault

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-31	View prescriptions	Logged in as patient	All prescriptions displayed with doctor name, diagnosis, medications	High	Pass

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-32	Download prescription as PDF	Click download on prescription	PDF generated and downloaded	High	Pass
TC-33	Upload medical document	Valid file	File uploaded to Supabase Storage, record saved	Medium	Pass
TC-34	Delete medical document	Existing document	Document deleted from storage and database	Low	Pass

Module 5: Medicine Reminders

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-35	Add medicine reminder with valid data	Medicine name, dosage, frequency, time, dates	Reminder created and displayed	Medium	Pass
TC-36	Mark medicine as taken	Active reminder	<code>taken: true,</code> <code>takenAt</code> timestamp saved	Medium	Pass
TC-37	Undo taken medicine	Taken reminder	<code>taken: false,</code> <code>takenAt</code> cleared	Medium	Pass
TC-38	Delete medicine reminder	Existing reminder	Reminder removed from list	Medium	Pass
TC-39	Automated email reminder	Cron job trigger	Email sent to patient for due reminders	Medium	Pass

Module 6: Admin

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-40	View admin dashboard	Logged in as admin	Real-time stats displayed (users, patients, doctors, appointments, prescriptions, contacts)	High	Pass
TC-41	View all users	Logged in as admin	All users listed with profiles and roles	High	Pass
TC-42	Delete another user	Valid user ID (not self)	User deleted from database	High	Pass

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-43	Delete own admin account	Admin's own user ID	Error: Cannot delete own account	High	Pass
TC-44	Filter appointments by status	Status = CONFIRMED	Only confirmed appointments displayed	Medium	Pass
TC-45	View contact messages	Logged in as admin	All contact form submissions displayed	Low	Pass

Module 7: Settings

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-46	Update profile name	New valid name	Name updated in database and session	Medium	Pass
TC-47	Change password with correct current password	Valid current + new password	Password updated successfully	Medium	Pass
TC-48	Change password with wrong current password	Invalid current password	Error: "Current password is incorrect"	Medium	Pass
TC-49	Change password with mismatched new passwords	New $\neq$ confirm password	Error: "Passwords do not match"	Medium	Pass
TC-50	Toggle 2FA on	2FA currently disabled	<code>twoFactorEnabled: true</code> saved	Medium	Pass
TC-51	Toggle 2FA off	2FA currently enabled	<code>twoFactorEnabled: false</code> saved	Medium	Pass

Module 8: Security Testing

TC ID	Test Case	Input	Expected Output	Priority	Status
TC-52	Access protected API without session	GET <code>/api/appointment</code> without auth	401 Unauthorized	High	Pass
TC-53	Access admin API as patient	GET <code>/api/admin/stats</code> as patient	403 Forbidden	High	Pass
TC-54	Access admin API as doctor	DELETE <code>/api/admin/users/[id]</code> as	403 Forbidden	High	Pass

TC ID	Test Case	Input	Expected Output	Priority	Status
doctor					
TC-55	SQL injection attempt	' OR 1=1 -- in login email field	Prisma rejects, no data exposed	High	Pass
TC-56	Access another patient's data	Patient A accessing Patient B's appointments	Only own data returned	High	Pass

Task 3: Requirement Traceability Matrix (RTM)

FR ID	Requirement	Test Case(s)	Status
FR-01	User Registration	TC-01, TC-02, TC-03, TC-04, TC-05	Pass
FR-02	Email Verification	TC-06, TC-07	Pass
FR-03	Credential Login	TC-08, TC-09, TC-10	Pass
FR-04	OAuth Login	TC-11, TC-12	Pass
FR-05	2FA via Email OTP	TC-13, TC-14, TC-15, TC-16, TC-17, TC-18, TC-19	Pass
FR-06	Book Appointment	TC-20, TC-21	Pass
FR-07	Manage Appointments	TC-24, TC-25, TC-26, TC-27	Pass
FR-08	Issue Prescription	TC-28, TC-29, TC-30	Pass
FR-09	View/Download Prescription	TC-31, TC-32	Pass
FR-10	Medicine Reminders	TC-35, TC-36, TC-37, TC-38	Pass
FR-11	Medical Vault Upload	TC-33	Pass
FR-12	Delete Medical Report	TC-34	Pass
FR-13	Profile/Password Update	TC-46, TC-47, TC-48, TC-49	Pass
FR-14	2FA Toggle in Settings	TC-50, TC-51	Pass
FR-15	Admin Dashboard Stats	TC-40	Pass
FR-16	Admin User Management	TC-41, TC-42, TC-43	Pass
FR-17	Admin Appointment Monitor	TC-44	Pass
FR-18	Admin Contact Messages	TC-45	Pass
FR-20	Automated Reminders (Cron)	TC-39	Pass
NFR-08	RBAC on APIs	TC-52, TC-53, TC-54	Pass
NFR-13	Input Validation	TC-55	Pass

Key Findings / Learning Outcomes

- Designed **56 test cases** covering all 8 modules of MediScript-E
- Understood the difference between **verification** (correct implementation) and **validation** (correct system)
- Learned that security test cases are as important as functional test cases in healthcare applications
- RTM ensures every functional requirement has at least one corresponding test case — no requirement is untested
- Recognized that **edge cases** (expired tokens, wrong OTP, self-deletion prevention) are critical to test
- Testing revealed that **Prisma ORM** effectively prevents SQL injection without additional configuration